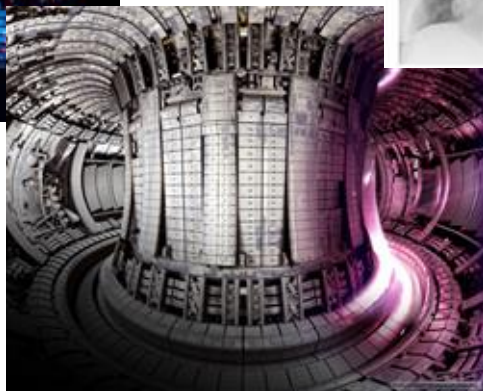
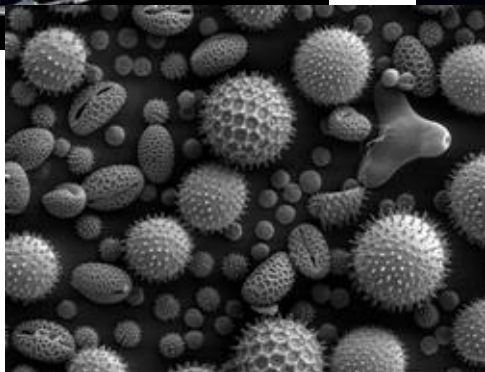
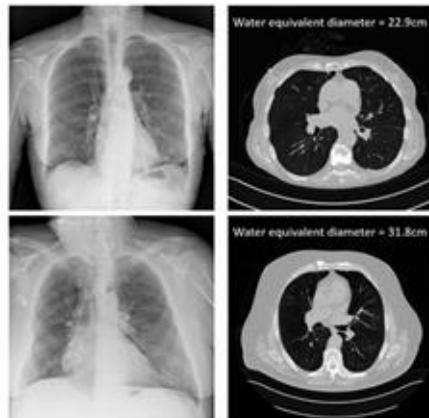
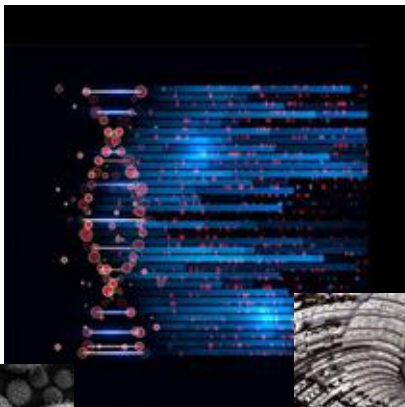
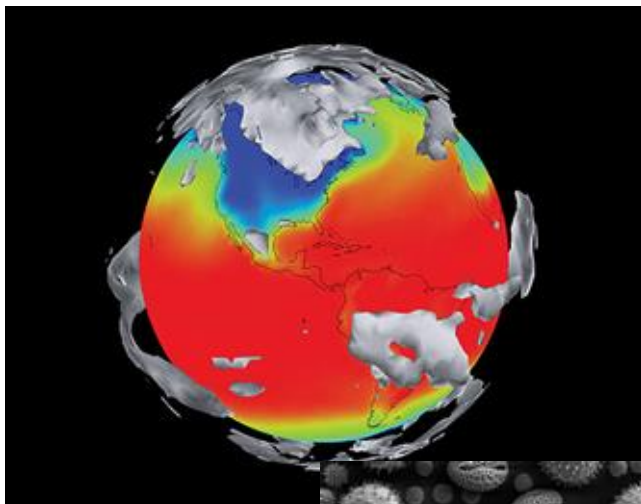


Intro to Big Array Data Analysis in Python

for geospatial science and microscopy

Multidimensional array data is everywhere



and more

What is Xarray?

Xarray is a cross-domain Python package that enables the analysis of large multidimensional datasets (mainly within the scientific domain). It does this in a few key ways:

- **introduces labels** in the form of **dimensions, coordinates, and attributes** on top of raw NumPy-like arrays, which allows for more intuitive, more concise, and less error-prone user experience
- **Integrates with lazy-array backends** (such as Dask) – allowing for the planning and execution of operations without exceeding RAM limitations

Let's dive into Xarray!



The website for this workshop



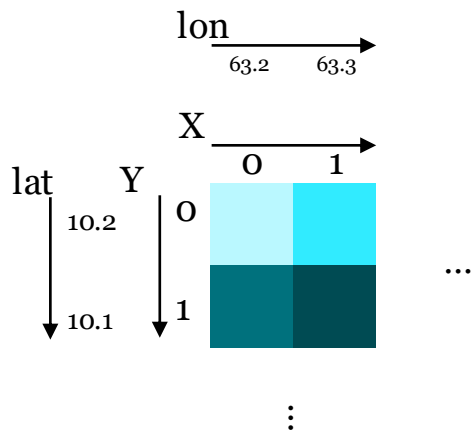
<https://uu-imau.github.io/xarray-UNECSS-2026/>

Tensors vs. tables

A fundamental distinction

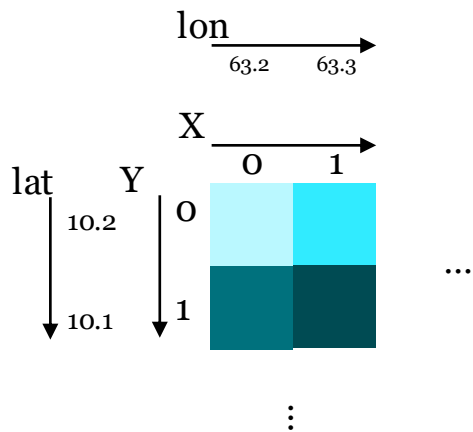
Tensors vs. tables

A fundamental distinction



Tensors vs. tables

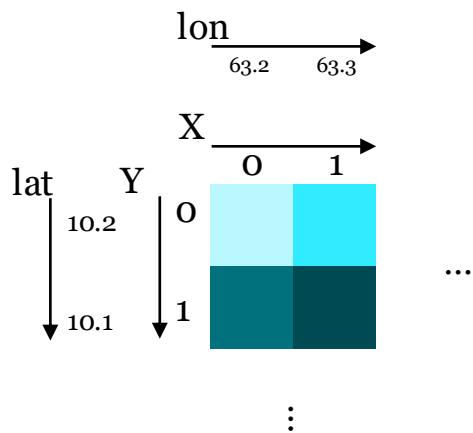
A fundamental distinction



X	Y	lon	lat	value

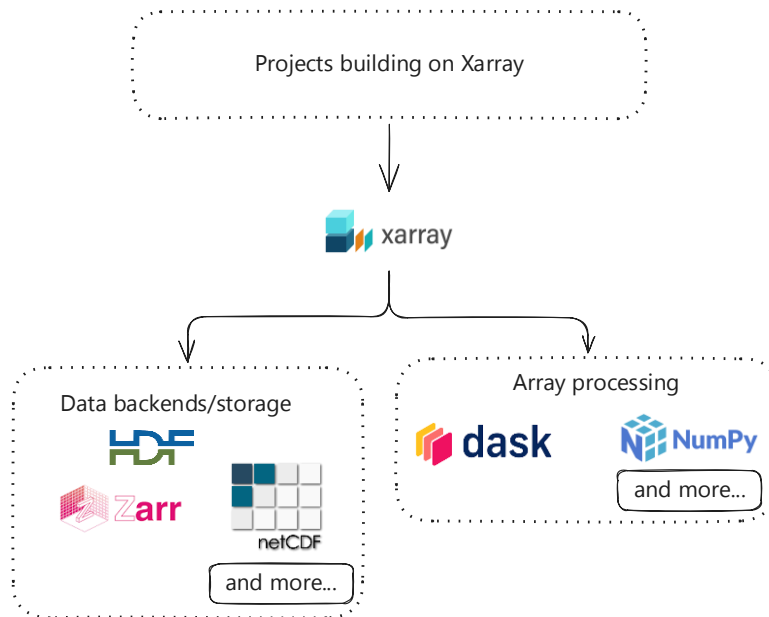
Tensors vs. tables

A fundamental distinction



X	Y	lon	lat	value
0	0	63.2	10.2	
1	0	63.3	10.2	
0	1	63.2	10.1	
1	1	63.3	10.1	

The ecosystem at a glance



Maturity of Xarray in various domains

- **Geosciences:** Very mature with lots of documentation, and good support for geoscience specific tools. The [Pythia Foundations course](#) from Project Pythia is an amazing resource for learning Python-based geoscientific workflows (it covers everything you need to know – and is my go-to recommendation for geoscientists). The Xarray documentation itself also have geoscience examples.
- **Microscopy:** Emerging support. [Napari](#) is rewriting their internals to use Xarray. Initiatives such as the [Open Microscopy Environment \(OME\)](#) are pushing the frontier on microscopy data in the cloud, and integration with Xarray. Currently, this intersection* suffers from poor documentation and tutorial resources – a gap that is being actively worked on (join the [Xarray Zulip](#) and chat with Ian Hunt-Isaak if you're interested here!).
- **Other domains:** The intersections* are not well defined or documented.

* i.e., the intersection between Xarray and using domain-specific file formats, and implementing domain specific workflows in Xarray

A new frontier in the cloud